Un sistem este reprezentat prin:

O unică funcție de transfer $H(s)$ ⟷ O multitudine de realizări în spațiul stărilor.

$(A_1, B_1, C_1, \Delta_1)$

$(A_2, B_2, C_2, \Delta_2)$

$\vdots$

$(A_x, B_x, C_x, \Delta_x)$

Pr. Se dă sistemul cu funcția de transfer:

$$H(s) = \frac{s+1}{(s+3)(s+2)}$$

Se cere:

a) Structura sistemului

b) O repr. în SS care să pună în evidență coeficienții polinomului caracteristic (de la numitorul funcției de transfer)

c) O repr. în SS. care să pună în evidență polii sistemului

d) Să se verifice ⟷ controlabilitate, observabilitate ⟷ stabilitate ⟷ internă, externă

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a) $H(s) = \frac{s+1}{(s+3)(s+2)} = \frac{s+1}{s^2+5s+6}$

* Ordin sistem = 2 (grad polinom numitor)

⇒ poli sistem ⟷ -3, -2. ⇒ 2 variabile de stare $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

⇒ dim $A = 2 \times 2$

* Pt că $H(s)$ este un raport de polinoame

⇒ 1 intrare ⇒ B are 1 coloană cu 2 elem.

⇒ 1 ieșire ⇒ C are 1 linie cu 2 elem.

b) $H(s) = \frac{s+1}{s^2+5s+6} = \frac{Y(s)}{U(s)}$ $\xrightarrow{\mathcal{L}^{-1}}$ $\begin{cases} \dot{x} = A\cdot x + B\cdot u \quad (1) \\ y = C\cdot x + D\cdot u \quad (2) \end{cases}$

$\Rightarrow Y(s) = \frac{(s+1)U(s)}{s^2+5s+6}$ $\xrightarrow{\mathcal{L}^{-1}}$

$\mathcal{L}[\dot{x}] = sX(s)$

$\mathcal{L}[\ddot{x}] = s^2X(s)$

$\mathcal{L}[x] = X(s)$

Notam $Z(s) = \frac{U(s)}{s^2+5s+6} \Rightarrow Z(s)[s^2+5s+6] = U(s)$

$\xrightarrow{\mathcal{L}^{-1}} \boxed{\ddot{z} + 5\dot{z} + 6\cdot z = u}$

Notam $x_1 = z \Rightarrow \dot{x}_1 = \dot{z} = x_2$

$\dot{x}_2 = \ddot{z} = u - 5\dot{z} - 6z = u - 5\cdot x_2 - 6x_1$

$\dot{x}_1 = 0\cdot x_1 + 1\cdot x_2 + 0\cdot u$

(1): $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} [u]$

$Y(s) = (s+1)Z(s) \xrightarrow{\mathcal{L}^{-1}} y = \dot{z} + z = x_2 + x_1$

(2): $y = \underbrace{[1 \quad 1]}_{C} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{[0]}_{D}[u]$

c) $H(s) = \frac{s+1}{(s+3)(s+2)} = \frac{\frac{s+2}{a}}{s+3} + \frac{\frac{s+3}{b}\cdot}{s+2} = \frac{as+2a+bs+3b}{(s+3)(s+2)}$

$H(s) = 2\cdot\frac{1}{s+3} - \frac{1}{s+2} = \frac{Y(s)}{U(s)} = \frac{s(a+b)+2a+3b\cdot}{(s+3)(s+2)}$

$\Rightarrow Y(s) = 2\cdot\underbrace{\frac{U(s)}{s+3}}_{X_1(s)} - \underbrace{\frac{U(s)}{s+2}}_{X_2(s)}$

$1 = a+b = 1a = 1-b$

$1 = 2a+3b.$

$1 = 2-2b+3b \Rightarrow b = -1$

$a = 1+1 = 2$

Notăm:

$$X_1(s) = \frac{U(s)}{s+3} \Rightarrow (s+3)X_1(s) = U(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} \dot{x}_1 + 3x_1 = u \Rightarrow \boxed{\dot{x}_1 = -3x_1 + u}$$

$$X_2(s) = \frac{U(s)}{s+2} \Rightarrow (s+2)X_2(s) = U(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} \dot{x}_2 + 2x_2 = u \Rightarrow \boxed{\dot{x}_2 = -2x_2 + u}$$

$$(1): \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{B} [u]$$

$$Y(s) = 2 \cdot X_1(s) - X_2(s) \overset{\mathcal{L}^{-1}}{\Rightarrow} y = 2x_1 - x_2$$

$$(2): y = \underbrace{[2 \quad -1]}_{C} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{[0]}_{D}[u]$$

d) Controlabilitate

$$R_c = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B] \qquad n = 2 - \text{ordin sistem}$$

$$R_c = [B \quad AB] = \begin{bmatrix} 1 & -3 \\ 1 & -2 \end{bmatrix} \quad (B \;\; AB)$$

$$\begin{vmatrix} 1 & -3 \\ 1 & -2 \end{vmatrix} = -2 + 3 = 1 \neq 0$$

$\Rightarrow \operatorname{rang} R_c = 2 \Rightarrow$ sistem controlabil

Observabilitate:

$$R_o = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \qquad n = 2 \Rightarrow \quad R_o = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -6 & 2 \end{bmatrix} \begin{matrix} \} C \\ \} CA \end{matrix}$$

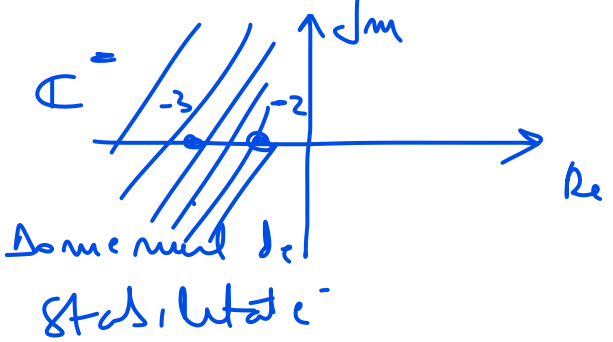
$$\begin{vmatrix} 2 & -1 \\ -6 & 2 \end{vmatrix} = 4 - 6 = -2 \neq 0 \Rightarrow \operatorname{rang} R_o = 2$$

$\Rightarrow$ sistem observabil

Stabilitate internă

$$[sI - A] = s \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} s+3 & 0 \\ 0 & s+2 \end{bmatrix}$$

$|sI-A| = (s+3)(s+2) = 0 \Rightarrow \Lambda(A) \in \{-3; -2\}$.

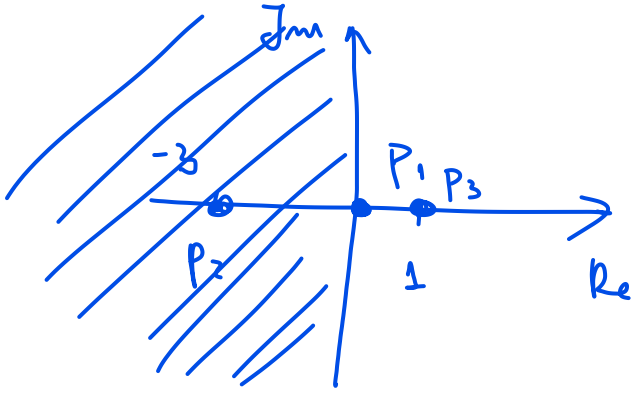


$\Rightarrow \Lambda(A) \subset \mathbb{C}^- \Rightarrow$ sistem stabil intern

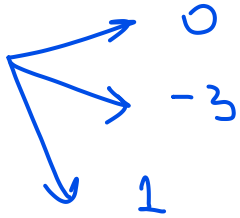
Stabilitate externa

$H(s) = \dfrac{s+1}{(s+3)(s+2)} \Rightarrow$ poli $\in \{-3; -2\} \subset \mathbb{C}^- \Rightarrow$ sistem stabil extern

Pr. 2 $H(s) = \dfrac{s+1}{s(s+3)(s-1)}$



ordin = 3 $\Rightarrow$ poli



$\Rightarrow$ poli $\notin \mathbb{C}^-$

$\Rightarrow$ sistem instabil extern